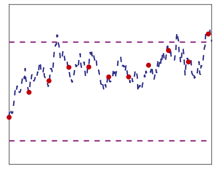
Chapter 12

Mean Exit Times



Outline

- Mean exit time
- Monte Carlo approach
- Computational experiments

12.1 Motivation

When does a particle first wander more than a set distance from its starting point? When does an interest rate first leave a specified band? When does the protein concentration in a cell exceed a critical threshold? Exit time problems like these are common in many fields. In this chapter, we use numerical SDE approximation in a Monte Carlo setting to compute the relevant *mean exit times*.

12.2 Background

Given the scalar SDE

$$\mathrm{d}X(t) = f(X(t))\,\mathrm{d}t + g(X(t))\,\mathrm{d}W(t), \quad X(0) = X_0, \tag{12.1}$$

where, for simplicity, we assume that X_0 is a constant lying in some interval (a, b), we may define the random variable T_{exit} to be

$$T_{\mathrm{exit}} := \inf\{t : X(t) = a \text{ or } X(t) = b\}.$$

In words, T_{exit} is the first time that the solution leaves the interval (a, b). Our task is then to find the mean exit time

$$T_{\mathrm{exit}}^{\mathrm{mean}} := \mathbb{E}\left[T_{\mathrm{exit}}\right].$$

12.3 Monte Carlo for Mean Exit Time

A straightforward Monte Carlo approach to approximate $T_{\mathrm{exit}}^{\mathrm{mean}}$ may be summarized as follows, where we use the EM method (8.3) to simulate the SDE (12.1).

Choose a stepsize, Δt
Choose a number of paths, M
for $s = 1$ to M
    Set $t_n = 0$ and $X_n = X_0$
    While $X_n > a$ and $X_n < b$
        Compute a $N(0,1)$ sample ξ_n
        Replace X_n by $X_n + \Delta t f(X_n) + \sqrt{\Delta t}\,\xi_n g(X_n)$
        Replace t_n by $t_n + \Delta t$
    end
    set $T^s_{\text{exit}} = t_n - \frac{1}{2}\Delta t$
end
set $a_M = \frac{1}{M}\sum_{s=1}^{M} T^s_{\text{exit}}$
set $b_M^2 = \frac{1}{M-1}\sum_{s=1}^{M}(T^s_{\text{exit}} - a_M)^2$

This produces an approximate mean exit time a_M and an approximate 95% confidence interval (2.6).

With this approach, we apply the numerical method on the discrete grid $\{t_i\}$ and, for each path, record the midpoint value $\frac{1}{2}(t_{n-1} + t_n)$, where t_n is the first gridpoint at which the numerical solution leaves the range (a, b). There are two main sources of error that we have already encountered in this book.

1. The sampling error arising from the approximation of an expected value by a sample mean.

2. The discretization error arising because we use a numerical method to approximate the SDE solution paths.[11]

However, there is also a third source of error, caused by the fact that we are recording solution values only at the discrete points $\{t_i\}$. Even if we follow the SDE solution paths exactly, we run the risk of missing an exit point—within an interval $t_i < t < t_{i+1}$ the path may leave the range (a, b) and then return unnoticed. Figure 12.1 illustrates this possibility.

Our aim here is to examine the accuracy of the Monte Carlo approach to estimating a mean exit time.

Under appropriate conditions, the mean exit time problem may also be formulated as a deterministic ODE. Letting $u(x)$ denote the mean exit time $T^{\text{mean}}_{\text{exit}}$ when $X_0 = x$, it can be shown that

$$\tfrac{1}{2}g(x)^2 \frac{\mathrm{d}^2 u}{\mathrm{d}x^2} + f(x)\frac{\mathrm{d}u}{\mathrm{d}x} = -1 \quad \text{for } a < x < b \text{ with } u(a) = u(b) = 0. \tag{12.2}$$

This gives an alternative approach for computing mean exit times: solve the ODE (12.2) either analytically or numerically. With a reference solution at hand we will be able to investigate the accuracy of the Monte Carlo approach.

12.4 Computational Experiments

First we consider the case where $f(X) = \mu X$ and $g(X) = \sigma X$, with $0 < a < X_0 < b$. We recall from Chapter 5 that this SDE is used to model asset prices, so the mean

[11] Moreover, the path simulation proceeds until X_n leaves the range (a, b), rather than over a fixed interval of time. Hence, the standard finite-time convergence theory, as outlined in Chapters 9 and 10, does not apply directly.

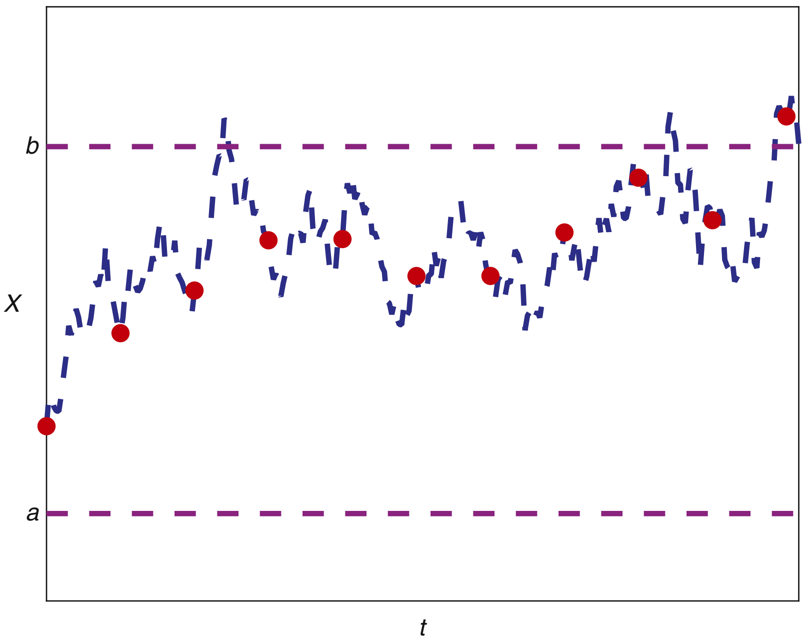


Figure 12.1. *A solution path (dashed curve) and values along the path at a discrete set of times (circles). The continuous-time path sneaks above the upper bound b twice without being spotted by the discrete samples. Only the third exit leads to a discrete value outside the range.*

exit time gives us the expected value of the first time that the asset either dips below the lower bound a or rises above the upper bound b. For this SDE, (12.2) becomes

$$\tfrac{1}{2}\sigma^2 x^2 \frac{\mathrm{d}^2 u}{\mathrm{d}x^2} + \mu x \frac{\mathrm{d}u}{\mathrm{d}x} = -1 \quad \text{for } a < x < b \text{ with } u(a) = u(b) = 0. \tag{12.3}$$

It can be shown that (12.3) has the solution

$$u(x) = \frac{1}{\tfrac{1}{2}\sigma^2 - \mu}\left(\log(x/a) - \frac{1-(x/a)^{1-2\mu/\sigma^2}}{1-(b/a)^{1-2\mu/\sigma^2}}\log(b/a)\right). \tag{12.4}$$

Exercise 12.1 asks you to confirm this, and Exercises 12.2–12.6 look at various properties of this formula.

Figure 12.2 illustrates the function $u(x)$ in the case where $\mu = 0.1$, $\sigma = 0.2$, $a = 0.5$, and $b = 2$. We see that the curve is convex with a fairly sharply defined maximum value around $x \approx 0.75$. This is intuitively reasonable. With an initial value x close to a, the first exit times will typically be close to zero (because the paths start very close to a boundary). As the initial value moves further away from a, the expected time to leave the range increases. At the other extreme, as the initial value approaches the upper limit b, the mean first exit time again tends to zero. The intermediate initial value $x \approx 0.75$, where $T_{\text{exit}}^{\text{mean}}$ is largest, is closer to a than to b. This asymmetry can be understood by looking at the exact solution (5.5) of the SDE and its expected value (5.6). We see that there is a general upward drift of solution paths, so, to get the maximum expected mean first exit time, the initial value should be closest to the lower bound.

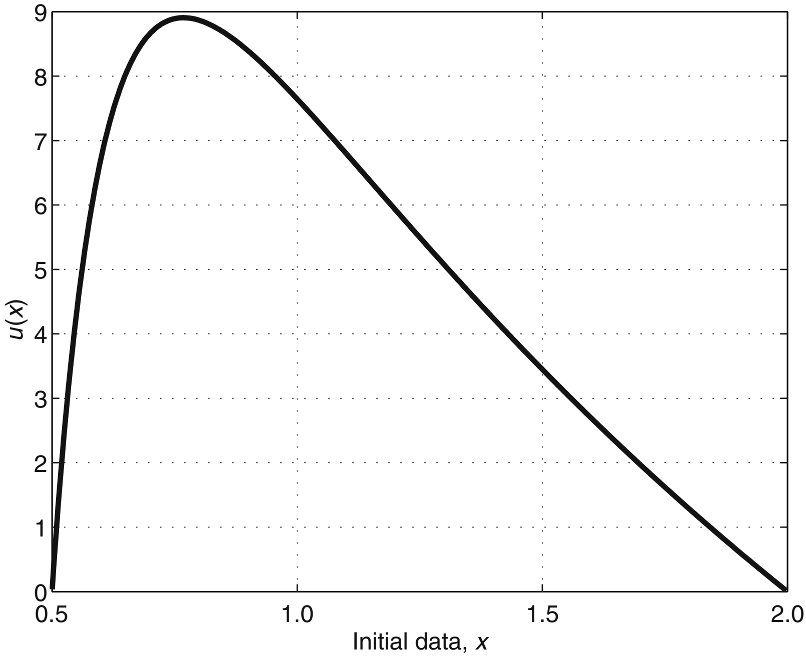


Figure 12.2. *The mean exit time function $u(x)$ in* (12.4).

Keeping $\mu = 0.1$, $\sigma = 0.2$, $a = 0.5$, $b = 2$, and taking $X_0 = 1$, we applied the Monte Carlo method with $M = 5 \times 10^4$ paths and a stepsize of $\Delta t = 10^{-2}$. For this SDE we know how the exact solution evolves, so instead of using the EM method we used the exact solution. That is, in the pseudocode description in Section 12.3, instead of

$$\text{Replace } X_n \text{ by } X_n + \Delta t \mu X_n + \sqrt{\Delta t}\, \xi_n \sigma X_n$$

we used

$$\text{Replace } X_n \text{ by } X_n \exp\left((\mu - \tfrac{1}{2}\sigma^2)\Delta t + \sqrt{\Delta t}\, \xi_n \sigma \right).$$

Figure 12.3 gives a histogram of the M exit times that arose. The sample mean for the exit times was 7.8056, with a sample variance of 5.6. This gives an approximate 95% confidence interval of $[7.7561, 7.8552]$. Using the formula (12.4), the exact value for the mean exit time is $T_{\text{exit}}^{\text{mean}} = 7.6450$. We conclude that

(a) the exact answer is well outside the confidence interval, and

(b) the Monte Carlo method overestimates the mean exit time.

Both features can be explained by arguing that the error from checking paths only at discrete time points $\{t_i\}$ is dominating the statistical sampling error. The confidence interval relates to the sample mean as an approximation to the mean exit time for the discrete process $\{t_i, X(t_i)\}$, but this quantity is a relatively poor overestimate of the mean exit time for the continuous-time process $\{t, X(t)\}$. This suggests that to improve the overall accuracy we should decrease the stepsize, Δt, rather than increase the number of sample paths, M. Reducing Δt to 10^{-3} produced a sample mean of 7.7137 with an approximate 95% confidence interval of $[7.6641, 7.7634]$. Further reducing Δt to 10^{-4} produced a sample mean of 7.6688 with an approximate

95% confidence interval of $[7.6200, 7.7177]$. These findings are consistent with the explanation above.

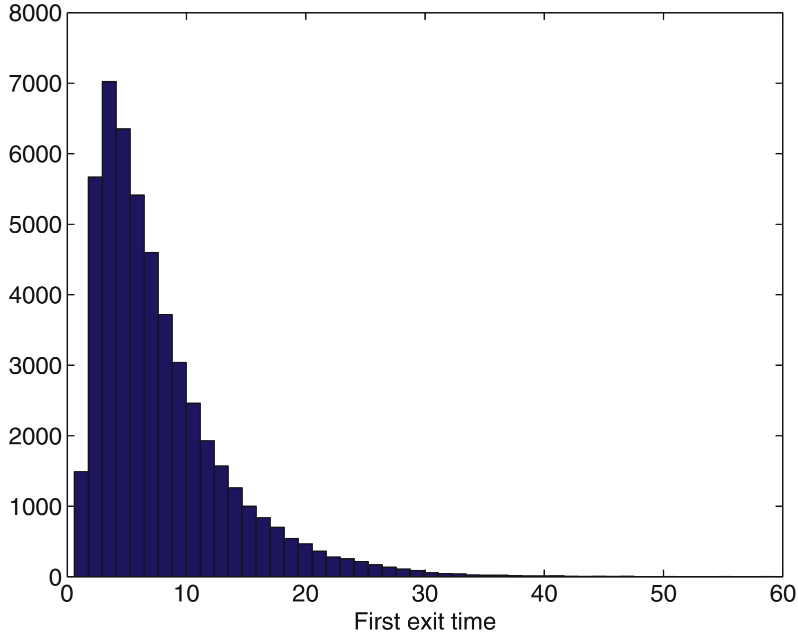


Figure 12.3. *Histogram of first exit times for $M = 5 \times 10^4$ sample paths.*

Next, we attempt to quantify the convergence rate of this method. Continuing with $\mu = 0.5$, $\sigma = 0.2$, $a = 0.5$, $b = 2$, and taking $X_0 = 1.5$, we fix the number of sample paths at $M = 5 \times 10^5$ and apply the algorithm with four different stepsizes: $\Delta t = 10^{-1}$, 10^{-2}, 10^{-3}, and 10^{-4}. The picture on the left of Figure 12.4 shows the sample means as crosses with confidence intervals as vertical bars. We have added the exact value, $T_{\text{exit}}^{\text{mean}} = 0.5993$, as a dashed line.

We may investigate this data further using a log–log plot and a least-squares fit, as described for Figure 8.2. Let us assume that the statistical sampling error is negligible and that the overall error, $\text{err}_{\Delta t} := |a_M - T_{\text{exit}}^{\text{mean}}|$, satisfies

$$\text{err}_{\Delta t} \approx C \Delta t^q \tag{12.5}$$

for some constants C and q. On the right of Figure 12.4 we plot $\text{err}_{\Delta t}$ as a function of Δt, using a log–log scale. A dashed reference line of slope $\frac{1}{2}$ is included. The error decay seems to be consistent with this slope. A least-squares fit to the power law $\log \text{err}_{\Delta t} = \log(C) + q \log(\Delta t)$ produced $q = 0.45$ with a residual of 0.1171.

This result is in agreement with the widely reported observation that the mean exit time based on a discrete grid gives an $O(\Delta t^{\frac{1}{2}})$ estimate of the mean exit time for the SDE; the result continues to hold when the EM method is used to approximate the SDE paths. See Section 12.5 for references.

Under this $O(\Delta t^{\frac{1}{2}})$ error behavior, Exercise 12.7 asks you to argue that to optimize computational effort the number of sample paths, M, should be of the order Δt^{-1}. In Figure 12.5 we applied the Monte Carlo method with $\Delta t = 10^{-4}$ and $M = 10^4$ for 40 equally spaced X_0 values in the range $[a, b]$. Here, we fixed $\mu = 0.1$ and $\sigma = 0.2$, with $a = 0.5$ and $b = 2$. The Monte Carlo sample means are plotted as crosses, with

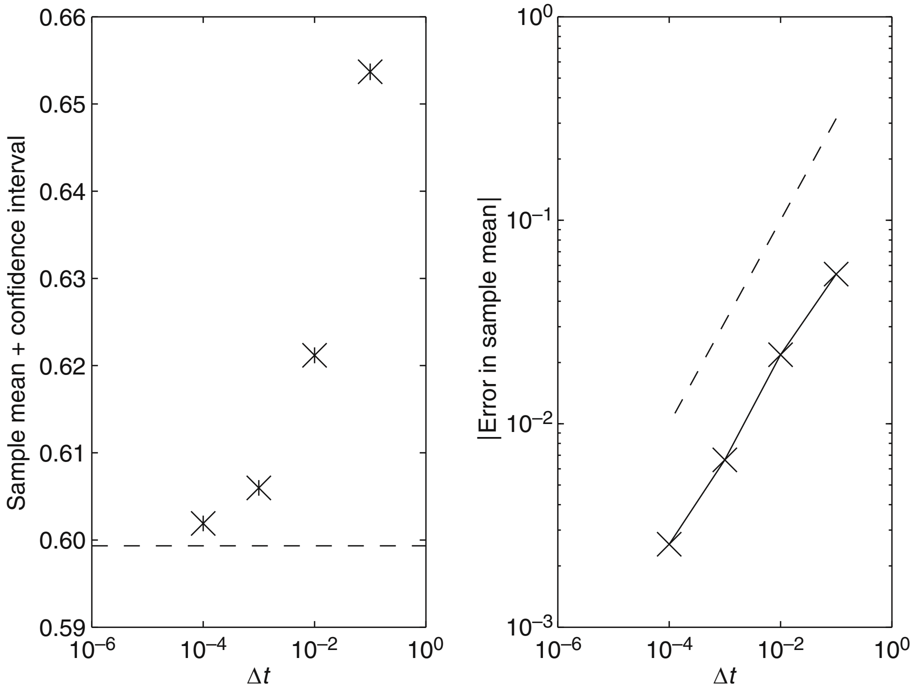


Figure 12.4. *Total error in Monte Carlo approximation to mean exit time for a range of stepsizes on a linear SDE.*

the corresponding 95% confidence intervals indicated by vertical lines. The exact mean exit times from (12.4) are plotted using a dashed line. With these parameter values the Monte Carlo method does a reasonable job of providing one or two digits of accuracy, typically being more accurate for X_0 close to the extremes of a and b, where the variance in the exit time is smaller.

For our next example, we repeat the type of experiment in Figure 12.5 for the case of the mean-reverting square root process (5.8). We recall that this SDE is used to model the movement of interest rates, and in this context the mean exit time measures, on average, the time at which the interest rate drifts outside a given range. We mentioned in Section 5.3 that, with $X_0 \geq 0$, the SDE solution is guaranteed to remain nonnegative. This property does not hold for a numerical method, in general. Hence, to avoid computing the square root of a negative number, we implement the EM method in the pseudocode of Section 12.3 as

$$\text{Replace } X_n \text{ by } X_n + \Delta t \lambda(\mu - X_n) + \sqrt{\Delta t}\, \xi_n \sigma \sqrt{|X_n|}. \tag{12.6}$$

In other words, we take the absolute value before applying the square root function.[12]

Setting $\lambda = 1$, $\mu = 0.5$, and $\sigma = 0.3$, with $a = 1$ and $b = 2$, we applied the Monte Carlo method with $M = 10^3$ sample paths and a stepsize of $\Delta t = 10^{-3}$. Results are shown in Figure 12.6. As in Figure 12.5, the sample means are shown as crosses, with 95% confidence intervals indicated by vertical lines, and the dashed line shows

[12] An alternative is to apply the standard EM method and then redefine X_n as $|X_n|$.

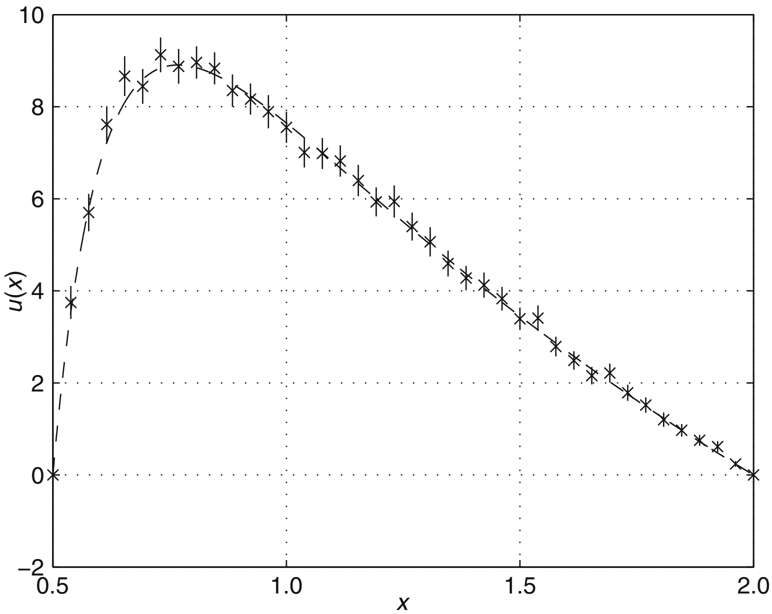


Figure 12.5. *Dashed line: mean exit time function $u(x)$ in (12.4). Crosses: Monte Carlo approximations, with confidence intervals indicated.*

the "exact" mean exit time curve. In this case, we used MATLAB's built-in function `bvp4c` to solve the relevant boundary-value ODE problem (12.2). We see that the accuracy of the EM-based Monte Carlo method is no worse than that in Figure 12.5, where the exact SDE solution was available and a smaller Δt and larger M were used.

As a final example, we consider the SDE (5.21) for a particle in a double-well potential with additive noise. We take $a = -3$ and $b = 1$. We recall from Section 5.3 that without noise, solutions must evolve in a way that lowers the potential. It is then interesting to see how the additive noise term allows solution paths to break this rule. Note that the potential function takes a much lower value at $x = b$ than at $x = a$, so, for small noise, an exit via b is much more likely than an exit via a. In other words, we are essentially measuring the mean time taken for the particle to climb over the hump and visit the right-hand well in Figure 5.3. In Figure 12.7 we show the mean exit time function $u(x)$ for three different noise levels—$\sigma = 2$ (solid), $\sigma = 4$ (dashed), and $\sigma = 6$ (dash-dotted)—computed from MATLAB's `bvp4c` routine. For $\sigma = 2$ we see that the mean exit time is very flat for initial values $-3 < x \leq -1$ other than those very close to -3. Here, only the upper limit $b = 1$ is relevant—despite starting near -3, solution paths rapidly drop down the potential well and are very likely to climb to the value $b = 1$ before $a = -3$. The function $u(x)$ thus has a *boundary layer* at the left-hand end. For $\sigma = 4$ and $\sigma = 6$ we are adding more noise, allowing more paths to reach $a = -3$ from a nearby starting point, which smooths out the boundary layer. Increasing σ also lowers all the mean exit times. Fixing $X_0 = 0$ and $\sigma = 4$, in Figure 12.8 we plot the overall error, $\mathrm{err}_{\Delta t} := |a_M - T_{\mathrm{exit}}^{\mathrm{mean}}|$, in the EM-based Monte Carlo method for $M = 5 \times 10^5$ paths and stepsizes of $\Delta t = 10^{-2}$, 10^{-3}, 10^{-4}, and 10^{-5}. As in Figure 12.4, we show the confidence intervals in the left-hand picture and give a log–log error plot in the right-hand picture. In this case, a least-squares fit gave a power of $q = 0.4561$ and a residual of 0.0174.

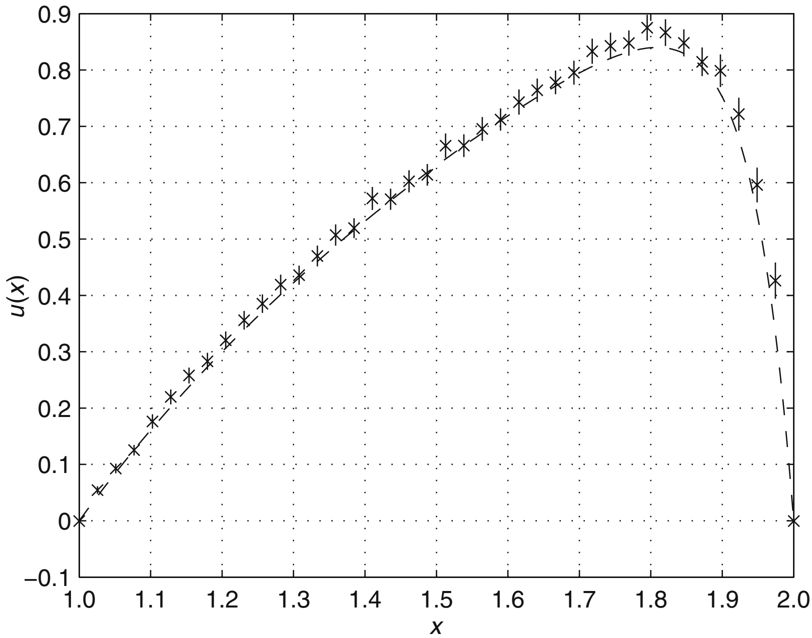


Figure 12.6. *Dashed line: mean exit time function $u(x)$ for square root process* (5.8)*. Crosses: Monte Carlo approximations, with confidence intervals indicated.*

12.5 Notes and References

The texts [56] and [127] offer accessible introductions to the theory of mean exit times, and in particular they show how the boundary-value ODE (12.2) arises. There are many interesting variations of this problem. For example, a boundary may *reflect* rather than *absorb*, or the quantity to be computed may be the probability that the solution reaches a before reaching b.

A more comprehensive numerical treatment of the double-well mean exit time problem can be found in [117].

Results concerning the *density* of the exit time for the mean-reverting square root process can be found in [141].

Several authors (see, for example, [18, 27, 117, 145]) have noted the $O(\Delta t^{\frac{1}{2}})$ error behavior in the type of simulations of this chapter. This convergence rate is disappointing: we saw in Chapter 9 that the EM method can approximate expected values of functions of the solution—quantities such as $\mathbb{E}[\Phi(X(T))]$—to an accuracy of $O(\Delta t)$. There are several ways to improve the accuracy of this basic algorithm.

- Adaptively reduce the stepsize when a path approaches a boundary.

- After each step, calculate the probability that an exit was missed. Then draw from a uniform $(0, 1)$ random number generator in order to decide whether to record an exit. Details for some simple SDEs can be found in [145].

- Use random stepsizes from a suitable exponential distribution, as described in [117, 118].

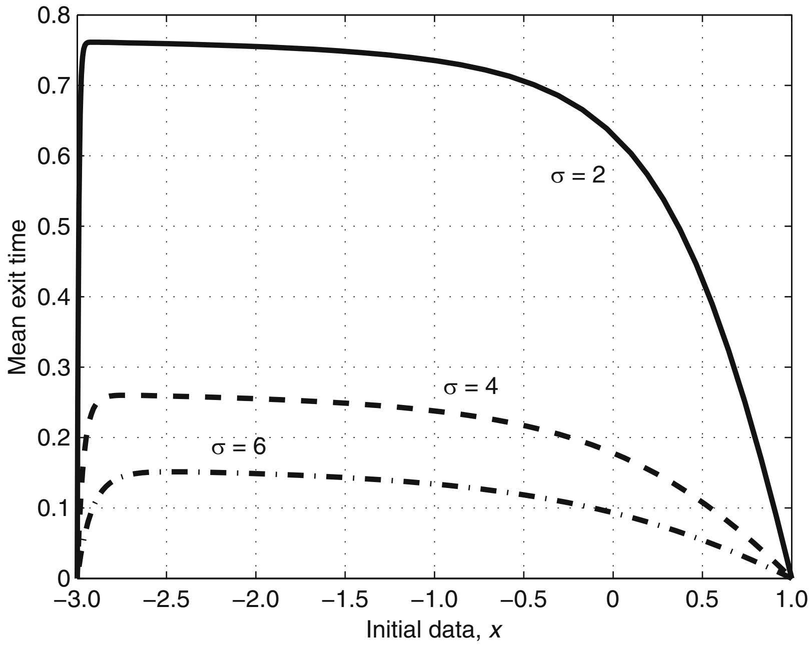


Figure 12.7. *The mean exit time function* $u(x)$ *from* (12.2) *for the double-well potential SDE* (5.21)*, with three different additive noise levels.*

We are not aware of a rigorous proof that, ignoring the statistical sampling issue, the error is $O(\Delta t^{\frac{1}{2}})$ for general computations along the lines of those in Figures 12.4 and 12.8. A key issue here is that, in principle, paths may take an arbitrarily long time to exit the specified domain, and hence convergence theory over a finite time $[0, T]$ is not directly applicable. (Convergence analysis for the related *killed diffusion* and *stopped diffusion* problems can be found, for example, in [73, 74, 75, 76].) On the other hand, we note that exit time problems are attractive in terms of analysis in the sense that the solution is restricted to a compact domain, and hence the traditional global Lipschitz assumption (5.22) on the drift and diffusion coefficients can be replaced by much less restrictive *local Lipschitz* versions: a suitable Lipschitz constant need only be found for the given interval (a, b).

Among the most efficient weak simulation algorithms for mean exit times are the random walk–based methods described in [157, Chapter 6]. See also Section 15.4 for reference to a *multilevel* Monte Carlo method. Of course, an alternative is to apply a numerical method to the deterministic ODE formulation (12.2), which in the case of a system of SDEs generalizes to a PDE. The stochastic simulation approach discussed here has the advantages that (a) it is straightforward to implement; (b) it can be used to find the mean exit time $u(x)$ for a single initial value, x, rather than the whole domain $a < x < b$; and (c) in the case of an SDE system it may not be feasible to discretize a PDE that has many spatial dimensions, or a complicated domain boundary.

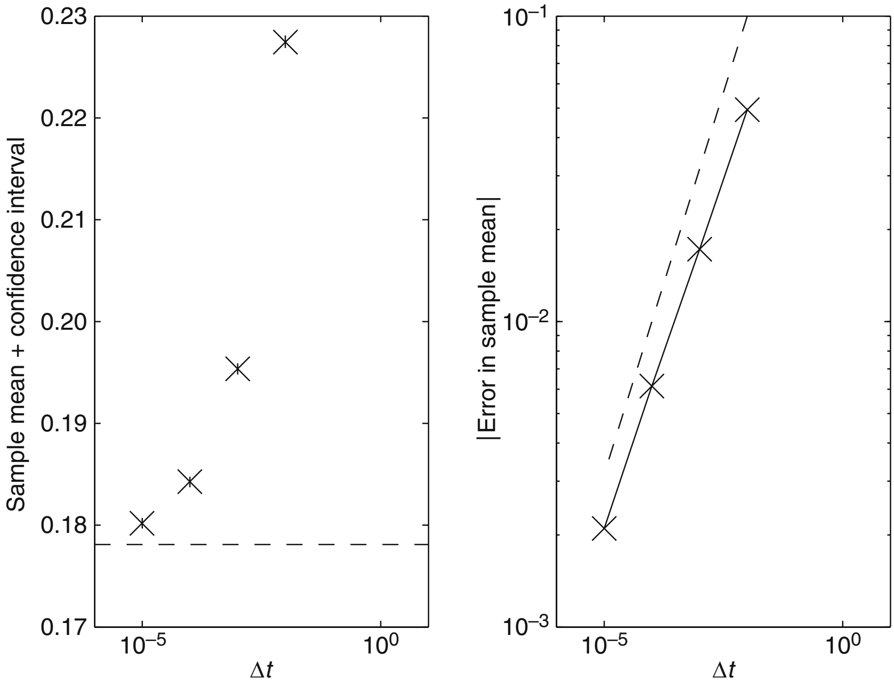


Figure 12.8. *Total error in Monte Carlo approximation to mean exit time for a range of stepsizes on a double-well potential SDE.*

Exercises

12.1. ⋆⋆ Confirm that (12.4) solves (12.3).

12.2. ⋆⋆ Show that there is a single point $x^\star$ over $[a, b]$ for which $u(x)$ in (12.4) is maximum, and find $u(x^\star)$.

12.3. ⋆ For $x = \frac{1}{2}(a + b)$, show that $u(x)$ in (12.4) satisfies $u(x) \to 0$ as $a \to b$ from below. Interpret this result.

12.4. ⋆⋆ Show that $u(x)$ in (12.4) satisfies $u(x) = C(x - a) + O(x - a)^2$ as $x - a$ tends to zero from above, and find the constant C.

12.5. ⋆⋆ Show that $u(x)$ in (12.4) satisfies $u(x) = D(b - x) + O(b - x)^2$ as $b - x$ tends to zero from above, and find the constant D.

12.6. ⋆ ⋆ ⋆ The expression for $u(x)$ in (12.4) appears to break down when $\mu = \frac{1}{2}\sigma^2$. By letting $\mu = \frac{1}{2}\sigma^2 + \epsilon$ and considering the limit $\epsilon \to 0$, show that a sensible mean exit time formula can be established in this special case.

12.7. ⋆ This exercise should be compared with Exercise 8.3. Referring to the Monte Carlo algorithm outlined in Section 12.3, suppose that the mean exit time for the discrete numerical solution differs from the mean exit time for the SDE by an amount that behaves like $O(\Delta t^{\frac{1}{2}})$ as $\Delta t \to 0$. Argue that the number of sample paths, M, used in the Monte Carlo method should be chosen so that M is proportional to Δt^{-1}.

12.8. ⋆⋆ The SDE $\mathrm{d}X(t) = \sqrt{X(t)\,(1 - X(t))}\,\mathrm{d}W(t)$ is a special case of a *Wright–Fisher diffusion model*, which arises in population dynamics; see, for example, [127,

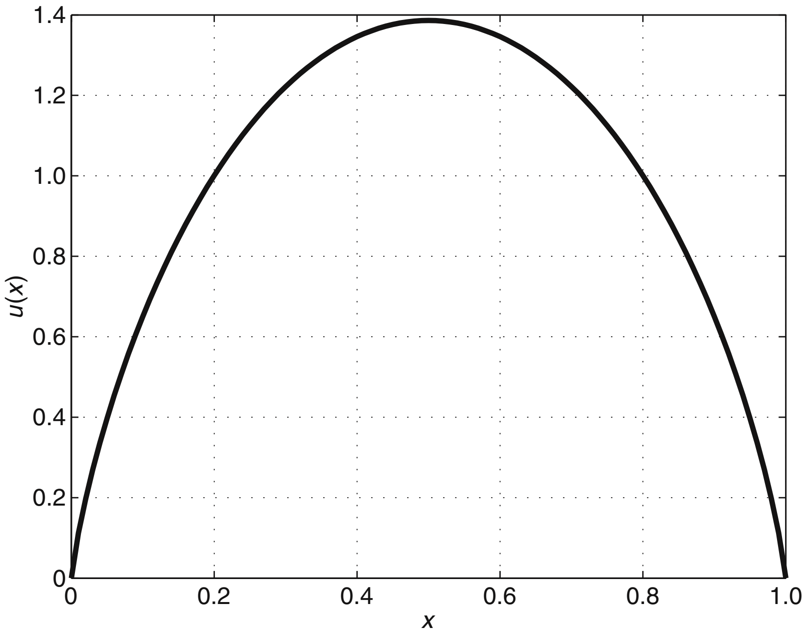


Figure 12.9. *The mean exit time function $u(x)$ for the SDE in Exercise* 12.8.

Section 12.2]. For $0 = a < X(0) < b = 1$, show that the ODE (12.2) for the mean exit time has solution $u(x) = -2\left((1-x)\log(1-x) + x\log x\right)$. This function is plotted in Figure 12.9. [Hint: recall that $\log x$ integrates to $x \log x - x$.]

12.9. $\star\star\star$ Consider the case where the SDE (12.1) is the mean-reverting square root process (5.8). Assume that $\mu = 1$ and $2\lambda = \sigma^2$. For given $0 < a < b$, find an analytical solution to the ODE (12.2) in terms of the *exponential integral* function $\mathrm{Ei}(x)$, which is defined as

$$\mathrm{Ei}(x) = \int_{-\infty}^{x} \frac{\mathrm{e}^t}{t}\, \mathrm{d}t.$$

12.10. $\star\star\star$ For $0 \le a < X(0) < b$, by considering the ODE (12.2), show that the mean exit time for the SDE

$$\mathrm{d}X(t) = k\,\mathrm{d}t + \sqrt{k}\,\mathrm{d}W(t),$$

where $k > 0$ is a constant, has the form

$$T_{\mathrm{exit}}^{\mathrm{mean}} = \frac{1}{k}\left[\frac{-\mathrm{e}^{-2x} + \mathrm{e}^{-2a}}{-\mathrm{e}^{-2b} + \mathrm{e}^{-2a}}\left(\frac{-\mathrm{e}^{-2b}}{2}\left(\mathrm{e}^{2b} - \mathrm{e}^{2x}\right) + b - x\right)\right.$$
$$\left. + \left(1 - \frac{-\mathrm{e}^{-2x} + \mathrm{e}^{-2a}}{-\mathrm{e}^{-2b} + \mathrm{e}^{-2a}}\right)\left(a - x + \frac{\mathrm{e}^{-2a}}{2}\left(\mathrm{e}^{2x} - \mathrm{e}^{2a}\right)\right)\right].$$

[Result is taken from [190].]

12.11. Repeat Exercise 12.10 for the SDE

$$\mathrm{d}X(t) = cX(t)\,\mathrm{d}t + \sqrt{c\,X(t)}\,\mathrm{d}W(t),$$

with $c > 0$ constant, to show that

$$T_{\text{exit}}^{\text{mean}} = \frac{1}{c}\left(\frac{e^{-2x}-e^{-2a}}{e^{-2b}-e^{-2a}}\left(-e^{-2b}\int_x^b \frac{e^{2l}}{l}\,dl + \ln b - \ln x\right)\right.$$
$$\left. + \left[1 - \frac{e^{-2x}-e^{-2a}}{e^{-2b}-e^{-2a}}\right]\left(\ln a - \ln x + e^{-2a}\int_a^x \frac{e^{2l}}{l}\,dl\right)\right),$$

where for the case $a = 0$ we may use $\lim \searrow 0$. [Result is also taken from [190].]

12.6 Program of Chapter 12 and Walk-through

The MATLAB code `ch12.m` in Listing 12.1 performs the type of exit time experiment described in Section 12.4, producing a picture like Figure 12.3. We simulate the linear SDE with parameters $\mu = 0.1$, $\sigma = 0.2$, $a = 0.5$, $b = 2$, and initial condition $X_0 = 1$. We apply the Monte Carlo method with $M = 10^5$ paths and a stepsize of $\Delta t = 2 \times 10^{-3}$. The program is based on the pseudocode in Section 12.4. Instead of using the EM method, we implement the exact path increment. This code produces a mean hitting time estimate of `tmean = 7.7297` with left and right confidence interval limits `cileft = 7.6946` and `ciright = 7.7648`. We also compute the exact solution from (12.4), `texact = 7.6450`. Finally, we display a histogram based on the 10^5 exit time samples, in order to illustrate the distribution of the underlying random variable.

Programming Exercises

P12.1. Extend `ch12.m` in order to obtain pictures like those in Figures 12.4, 12.5, 12.6, and 12.7.

P12.2. Experiment with the use of MATLAB's `bvp4c` for solving two-point boundary-value ODEs of the form (12.2) in order to obtain reference solutions for judging the accuracy of Monte Carlo simulations based on EM.

```
%CH12.M    Compute mean exit time using Monte Carlo
%
% Linear SDE with exact path simulation

clf
rng(500)
mu = 0.1; sigma = 0.2;                     % problem parameters
a = 0.5; b = 2; Xzero = 1;
dt = 2e-3; M = 1e5;                        % method parameters

texit = zeros(M,1);                        % pre-allocate array

for s = 1:M
     X = Xzero;
     t = 0;
     while X > a & X < b,
       dW = sqrt(dt)*randn;                           % Brownian inc
       X = X*exp((mu - 0.5*sigma^2)*dt + sigma*dW); % exact path
       t = t + dt;
     end
     texit(s) = t - 0.5*dt;
end

tmean = mean(texit)
tstd = std(texit);

cileft = tmean - 1.96*tstd/sqrt(M)
ciright = tmean + 1.96*tstd/sqrt(M)

%%% Compute exact value %%%

%%% Temporary variables to break up the formula %%%
temp1 = 1/(0.5*sigma^2 - mu);
temp2 = log(Xzero/a);
power = 1 - 2*mu/(sigma^2);
temp3 = 1 - (Xzero/a)^power;
temp4 = 1 - (b/a)^power;
temp5 = log(b/a);

%%% Mean Hitting time formula %%%
texact = temp1*( temp2 - (temp3/temp4)*temp5)

hist(texit,100)             % display a histogram
xlabel('First Exit Time')
```

Listing 12.1. *Program of Chapter* 12*:* *ch12.m.*

Quotes

Exit, pursued by a bear.

William Shakespeare (1564–1616), stage direction in *The Winter's Tale*.

At any given time, a large portion of the world's computers are doing Monte Carlo simulations.

Don Estep (during a lecture at the International Conference on Computational and Mathematical Methods in Science and Engineering, 2005).

Those who trust to chance must abide by the results of chance.

Calvin Coolidge (1872–1933) (source: `http://www.quotedb.com/quotes/3169`).